Parul® University
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NAAC GRADE A++

Information and Communication Technology

UNIT-1
Introduction

Study Guide

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UNIT – I INTRODUCTION

Data Science:

Data science is the domain of study that deals with vast volumes of data using modern tools and techniques, including essential data science skills, to find unseen patterns, derive meaningful information, and make business decisions. Data science uses complex machine learning algorithms to build predictive models. The data used for analysis can come from many different sources and presented in various formats. Now that you know what data science is, let's see the data science lifestyle.

- Uses of Data Science:

Data science may detect patterns in seemingly unstructured or unconnected data, allowing conclusions and predictions to be made.
Tech businesses that acquire user data can utilise strategies to transform that data into valuable or profitable information.
Data Science has also made inroads into the transportation industry, such as with driverless cars. It is simple to lower the number of accidents with the use of driverless cars. For example, with driverless cars, training data is supplied to the algorithm, and the data is examined using data Science approaches, such as the speed limit on the highway, busy streets, etc.
Data Science applications provide a better level of therapeutic customisation through genetics and genomics research.

Applications of Data Science

There are various applications of data science, including:

- Healthcare

  Healthcare companies are using data science to build sophisticated medical instruments to detect and cure diseases.

- Gaming

- Video and computer games are now being created with the help of data science and that has taken the gaming experience to the next level.

- Image Recognition
- Identifying patterns is one of the most commonly known applications of data science. in images and detecting objects in an image is one of the most popular data science applications.
- Recommendation Systems:Netflix and Amazon give movie and product recommendations based on what you like to watch, purchase, or browse on their platforms.
- Logistics

  Data Science is used by logistics companies to optimize routes to ensure faster delivery of products and increase operational efficiency.

- Fraud Detection
- Fraud detection comes the next in the list of applications of data science. Banking and financial institutions use data science and related algorithms to detect fraudulent transactions.
- Internet Search
- Speech recognition

  Speech recognition is one of the most commonly known applications of data science. It is a technology that enables a computer to recognize and transcribe spoken language into text. It has a wide range of applications, from virtual assistants and voice-controlled devices to automated customer service systems and transcription services.

- Targeted Advertising

  . From display banners on various websites to digital billboards at airports, data science algorithms are utilised to identify almost anything. This is why digital advertisements have a far higher CTR (Call-Through Rate) than traditional marketing. They can be customised based on a user's prior behaviour. That is why you may see adverts for Data Science Training Programs while another person sees an advertisement for clothes in the same region at the same time.

- Airline Route Planning
- Augmented Reality

Example of Data Science
Here are some brief examples of data science showing data science's versatility.

- Law Enforcement: In this scenario, data science is used to help police in Belgium to better understand where and when to deploy personnel to prevent crime. With only limited resources and a large area to cover data science used dashboards and reports to increase the officers' situational awareness, allowing a police force that's spread thin to maintain order and anticipate criminal activity.
- Pandemic Fighting: The state of Rhode Island wanted to reopen schools, but was naturally cautious, considering the ongoing COVID-19 pandemic. The state used data science to expedite case investigations and contact tracing, enabling a small staff to handle an overwhelming number of concerned calls from citizens. This information helped the state set up a call center and coordinate preventative measures.
- Driverless Vehicles: Lunewave, a sensor manufacturing company, was looking for a way to make sensor technology more cost-effective and accurate. They turned to data science and machine learning to train their sensors to be safer and more reliable, as well as using data to improve their 3D-printed sensor manufacturing process.

❖ The Data Science Lifecycle

1. Data science's lifecycle consists of five distinct stages, each with its own tasks:
2. Capture: Data Acquisition, Data Entry, Signal Reception, Data Extraction. This stage involves gathering raw structured and unstructured data.
3. Maintain: Data Warehousing, Data Cleansing, Data Staging, Data Processing, and Data Architecture. This stage covers taking the raw data and putting it in a form that can be used.
4. Process: Data Mining, Clustering/Classification, Data Modeling, Data Summarization. Data scientists take the prepared data and examine its patterns, ranges, and biases to determine how useful it will be in predictive analysis.
5. Analyze: Exploratory/Confirmatory, Predictive Analysis, Regression, Text Mining, and Qualitative Analysis. Here is the real meat of the lifecycle. This stage involves performing the various analyses on the data.
6. Communicate: Data Reporting, Data Visualization, Business Intelligence, and Decision Making. In this final step, analysts prepare the analyses in easily readable forms such as charts, graphs, and reports.

1.2. Digital Universe:

The term "digital universe" in data science refers to the entirety of digital data generated, stored, processed, and transmitted globally. This encompasses all types of data created by various sources such as individuals, organizations, and devices. The concept is significant in understanding the scope, scale, and impact of digital information on various aspects of society, economy, and technology.

key components and aspects of the digital universe in data science:

- Volume: The digital universe is characterized by its vast and exponentially growing volume of data. This includes data from social media, financial transactions, IoT devices, multimedia content, and more.
- Variety: Data in the digital universe comes in multiple formats and types, including structured data (databases, spreadsheets), unstructured data (text, images, videos), and semi-structured data (JSON, XML).
- Velocity: The speed at which data is generated and processed is a critical aspect. Real-time data from sensors, social media feeds, and financial markets exemplifies high-velocity data.
- Veracity: Ensuring the accuracy and reliability of data is crucial. The digital universe contains both high-quality data and significant amounts of noise and misinformation.
- Value: The ultimate goal of analyzing the digital universe is to extract valuable insights that can inform decision-making, drive innovation, and create competitive advantages.
- Big Data Technologies: The digital universe necessitates the use of advanced technologies and tools for storage, processing, and analysis. These include distributed computing frameworks (Hadoop, Spark), data warehouses, cloud storage solutions, and machine learning algorithms.
- Data Sources: The digital universe is populated by a diverse array of data sources, including social media platforms, sensors (IoT), mobile devices, enterprise systems, scientific research, and government databases.
- Privacy and Security: Managing the digital universe involves addressing significant challenges related to data privacy, security, and compliance with regulations like GDPR and CCPA.

❖ **Uses of Digital Universe:**

- The concept of the digital universe is essential in data science due to its expansive and diverse data landscape, which provides numerous opportunities for analysis, insights, and decision-making across various fields. Here are the key uses of the digital universe in data science:

- Insight Generation

- Data scientists analyze vast amounts of data from the digital universe to uncover patterns, trends, and correlations. This can lead to valuable insights for businesses, healthcare, finance, and more. For instance, analyzing social media data can reveal consumer sentiment and market trends.

- Predictive Analytics

  Predictive models built on data from the digital universe help forecast future events or behaviors. Retailers, for example, use predictive analytics to anticipate inventory needs based on past sales data, while healthcare providers predict patient outcomes from medical records.

- Personalization

  Data from the digital universe allows for highly personalized experiences. Online services like streaming platforms, e-commerce websites, and social media use data science to tailor recommendations and advertisements to individual user preferences and behaviors.

- Operational Efficiency

  Organizations leverage data from the digital universe to streamline operations and improve efficiency. Manufacturing processes can be optimized using IoT data, while logistics companies use real-time data for route optimization and fleet management.

- Fraud Detection and Security

  Financial institutions and online platforms use data science to detect fraudulent activities by analyzing transaction patterns and user behavior. Security systems can also monitor and analyze data traffic to identify and mitigate cyber threats.

- Healthcare Advancements

  The digital universe provides critical data for medical research and patient care. Analyzing electronic health records, medical imaging, and genomic data helps in disease diagnosis, treatment planning, and personalized medicine.

- Scientific Research

  Researchers across various fields use data from the digital universe to advance scientific knowledge. Environmental scientists, for example, analyze satellite data to monitor climate change, while astronomers use data from telescopes to study the universe.

- Business Intelligence

  Businesses use data from the digital universe for strategic planning and decision-making. Data science tools analyze sales data, customer feedback, and market trends to guide business strategies and improve competitiveness.

- Urban Planning and Smart Cities

  Data from sensors, mobile devices, and public services in the digital universe is used to enhance urban planning and develop smart cities. This includes traffic management, energy distribution, waste management, and public safety.

- Educational Insights

  Educational institutions analyze data from the digital universe to improve learning outcomes. This includes student performance data, engagement metrics, and feedback to personalize education and identify areas for improvement.

- Economic and Social Policy

  Governments and organizations use data science to inform policy decisions. Economic data analysis helps in understanding market dynamics and labor trends, while social data analysis aids in addressing public health issues and resource allocation.

- Sustainability and Environmental Monitoring: Data from the digital universe helps monitor and manage environmental resources. This includes tracking deforestation, pollution levels, and wildlife populations to promote sustainable practices and environmental conservation.
- Aspects of the Digital Universe in Data Science
- Big Data Technologies
- Tools and Frameworks: Technologies such as Hadoop, Spark, and NoSQL databases enable the processing and analysis of large datasets.

- Cloud Computing: Cloud platforms like AWS, Google Cloud, and Azure provide scalable resources for managing big data workloads.
- Data Analytics
- Descriptive Analytics: Summarizing historical data to understand what has happened.
- Predictive Analytics: Using statistical models and machine learning to predict future outcomes.
- Prescriptive Analytics: Recommending actions based on data insights to achieve desired outcomes.
- Machine Learning and AI: Algorithms: Applying machine learning algorithms to analyze patterns, classify data, and make predictions.
- Deep Learning: Leveraging neural networks for complex tasks such as image recognition, natural language processing, and autonomous systems.
- Data Governance
- Policies and Standards: Establishing frameworks for data management, privacy, and security.
- Compliance: Adhering to regulations like GDPR, CCPA, and industry-specific standards.
- Data Visualization
- Techniques: Creating visual representations of data to make insights more accessible and understandable.
- Tools: Utilizing tools like Tableau, Power BI, and D3.js to develop interactive and informative visualizations.
- Impact on Society
- Economic: Driving innovation, improving efficiency, and creating new business opportunities.
- Social: Enhancing quality of life through smart cities, personalized healthcare, and improved education.
- Ethical: Addressing concerns related to data privacy, security, and the ethical use of data.

Sources of Data

A data source may be the initial location where data is born or where physical information is first digitized, however even the most refined data may serve as a source, as long as another process accesses and utilizes it. Concretely, a data source may be a database, a flat file, live measurements from physical devices, scraped web data, or any of the myriad static and streaming data services which abound across the internet.

Data source types

Though the diversity of content, format, and location for data is only increasing with contributions from technologies such as IoT and the adoption of big data methodologies, it remains possible to classify most data sources into two broad categories: machine data sources and file date sources.

Though both share the same basic purpose — pointing to the data's location and describing similar connection characteristics — machine and file data sources are stored, accessed, and used in different ways.

Machine data sources

Machine data sources have names defined by users, must reside on the machine that is ingesting data, and cannot be easily shared. Like other data sources, machine data sources provide all the information necessary to connect to data, such as relevant software drivers and a driver manager, but users need only ever refer to the DSN as shorthand to invoke the connection or query the data.

The connection information is stored in environment variables, database configuration options, or a location internal to the machine or application being used. An Oracle data source, for example, will contain a server location for accessing the remote DBMS, information about which drivers to use, the driver engine, and any other relevant parts of a typical connection string, such as system and user IDs and authentication.

File data sources

File data sources contain all of the connection information inside a single, shareable, computer file (typically with a .dsn extension). Users do not decide which name is assigned to file data sources, as these sources are not registered to individual applications, systems, or users, and in fact do not have a DSN like that of machine data sources. Each file stores a connection string for a single data source.

File data sources, unlike machine sources, are editable and copyable like any other computer file. This allows users and systems to share a common connection (by moving the data source between individual machines or servers), and for the streamlining of data connection processes (for example by keeping a source file on a shared resource so it may be used simultaneously by multiple applications and users).

It is important to note that 'unshareable' .dsn files also exist. These are the same type of file as described above, but they exist on a single machine and cannot be moved or copied. These files point directly to machine data sources. This means that unshareable file data sources are wrappers for machine data sources, serving as a proxy for applications which expect only files but also need to connect to machine data.

How data sources work

Data sources are used in a variety of ways. Data can be transported thanks to diverse network protocols, such as the well-known File Transfer Protocol (FTP) and HyperText Transfer Protocol (HTTP), or any of the myriad Application Programming Interfaces (APIs) provided by websites, networked applications, and other services.

Many platforms use data sources with FTP addresses to specify the location of data needed to be imported. For example, in the Adobe Analytics platform, a file data source is uploaded to a server using an FTP client, then a service utilizes this source to move and process the relevant data automatically.

SFTP (The S stands for Secure or SSH) is used when usernames and passwords need to be obfuscated and content encrypted, or FTPS may alternatively be used by adding Transport Layer Security (TLS) to FTP, achieving the same goal.

Meanwhile, many and diverse APIs are now provided to manage data sources and how they are used in applications. APIs are used to programmatically link applications to data sources, and typically provide more customization and a more versatile collection of access methods. For example, Spark provides an API with abstract implementations for representing and connecting to data sources, from barebones but extensible classes for generic relational sources, to detailed implementations for hard-coded JDBC connections.

Other protocols for moving data from sources to destinations, especially on the web, include NFS, SMB, SOAP, REST, and WebDAV. These protocols are often used within APIs (and some APIs themselves make use of other APIs internally), within fully featured data applications, or as standalone transfer processes. Each have characteristic features and security concerns which should be considered for any data transfer.

The purpose of a data source

Ultimately, data sources are intended to help users and applications connect to and move data to where it needs to be. They gather relevant technical information in one place and hide it so data consumers can focus on processing and identify how to best utilize their data.

The purpose here is to package connection information in a more easily understood and user-friendly format. This makes data sources critical for more easily integrating disparate systems, as they save shareholders from the need to deal with and troubleshoot complex but low-level connection information.

And although this connection information is hidden, it is always accessible when necessary. Additionally, this information is stored in consistent locations and formats which can ease other processes such as migrations or planned system structural changes.

Information Commons:

"Information Commons" refers to shared resources and collaborative environments that facilitate access to and use of information and data for research, education, and innovation. Information Commons are designed to provide the infrastructure and support needed for data-driven work, fostering a culture of openness, collaboration, and knowledge sharing.

Key Components of an Information Commons in Data Science

- Data Repositories:Centralized databases where datasets are stored, cataloged, and made accessible to researchers and practitioners.
- Examples include public datasets like those from the U.S. Census Bureau, scientific datasets like those from the Human Genome Project, or open data portals from city governments.

Computing ResourcEe: High-performance computing (HPC) facilities, cloud computing services, and data storage solutions that provide the necessary computational power and storage capacity for data-intensive tasks.

- Resources such as AWS, Google Cloud Platform, Microsoft Azure, or institution-specific HPC cluster.

- **Collaborative Workspaces**

- Physical or virtual spaces designed for collaborative work, often equipped with advanced tools and technologies to support data analysis, visualization, and research collaboration.
- Examples include university-based data science labs, coworking spaces for tech startups, and virtual collaboration platforms like Jupyter Notebooks and GitHub.

- **Software and Tools**

- Access to a variety of software tools and platforms essential for data science work, such as statistical analysis software (R, SAS), machine learning frameworks (TensorFlow, PyTorch), and data visualization tools (Tableau, D3.js).
- Licensing agreements and training sessions to facilitate the use of these tools.

- **Knowledge Repositories**

- Libraries and digital archives that provide access to research papers, technical reports, case studies, and other scholarly resources.
- Examples include digital libraries like IEEE Xplore, arXiv, PubMed, and institutional repositories.

- **Educational and Training Programs**

Workshops, courses, seminars, and online tutorials aimed at building skills in data science, programming, statistical analysis, machine learning, and other relevant areas.
Partnerships with educational platforms like Coursera, edX, DataCamp, and in-house training programs.

  - Data Governance and Ethics Frameworks

Policies and guidelines to ensure the responsible use and sharing of data, addressing issues such as data privacy, security, intellectual property, and ethical considerations.
Compliance with regulations like GDPR, CCPA, and institutional review boards (IRBs).

- **Community and Networking:** Support for building and maintaining a community of data scientists, researchers, students, and professionals through events, conferences, hackathons, and online forums.

- Examples include meetups, professional associations like the Data Science Association, and online communities such as Kaggle and Stack Overflow.

- **Benefits of Information Commons in Data Science**

- Enhanced Collaboration: By providing shared resources and collaborative spaces, Information Commons foster interdisciplinary collaboration and collective problem-solving.

- Increased Access to Data and Tools: Making data and computational tools widely accessible democratizes the ability to perform advanced data science tasks, enabling more people to contribute to scientific and technological advancements.

- Accelerated Research and Innovation: Access to high-quality data and computational resources speeds up the research process, leading to faster innovation and discovery.

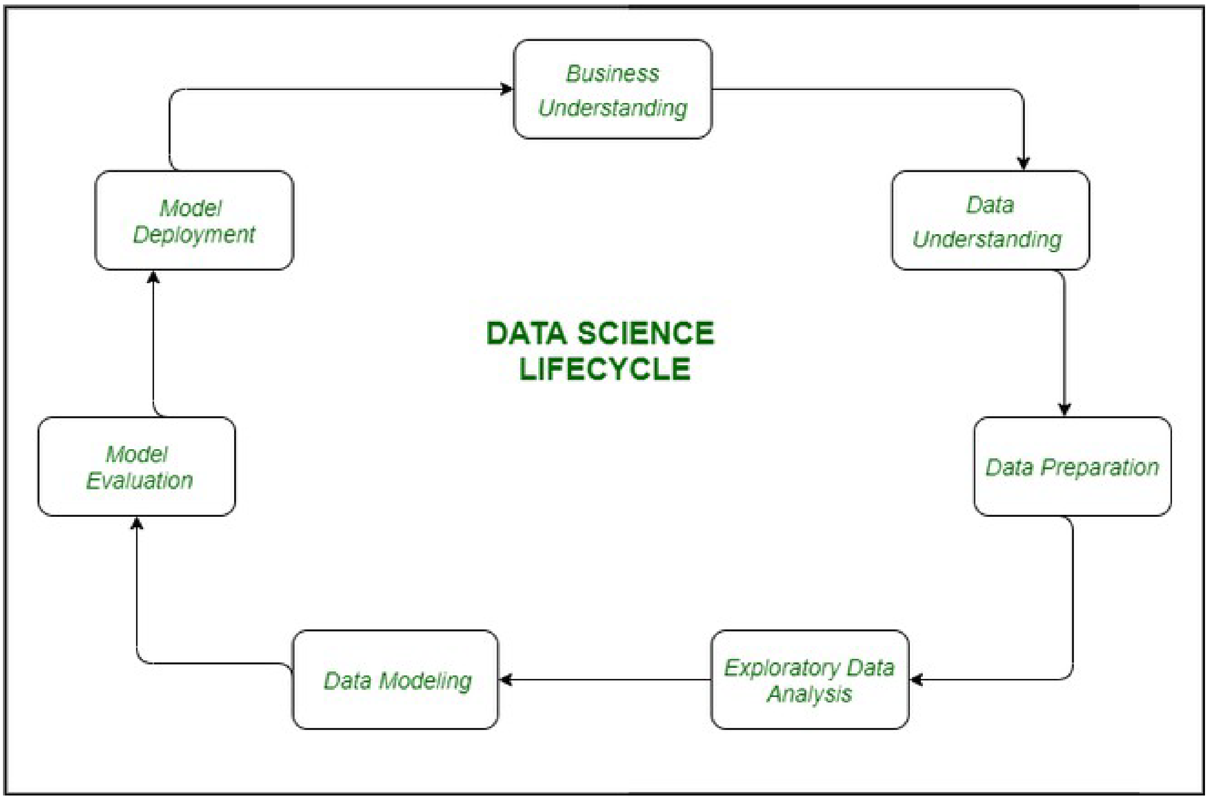
- Skill Development: Educational programs and resources help individuals build and enhance their data science skills, contributing to a more knowledgeable and skilled workforce.

- Promotion of Open Science: Information Commons support the principles of open science by promoting transparency, reproducibility, and the sharing of data and findings.

- Support for Data-Driven Decision Making: By providing robust data and analytical tools, Information Commons help organizations make more informed, evidence-based decisions.

Data Science Project Life Cycle

Data Science Lifecycle revolves around the use of machine learning and different analytical strategies to produce insights and predictions from information in order to acquire a commercial enterprise objective. The complete method includes a number of steps like data cleaning, preparation, modelling, model evaluation, etc. It is a lengthy procedure and may additionally take quite a few months to complete. So, it is very essential to have a generic structure to observe for each and every hassle at hand



1. **The lifecycle of Data Science**

2. Business Understanding: The complete cycle revolves around the enterprise goal. What will you resolve if you do not longer have a specific problem? It is extraordinarily essential to apprehend the commercial enterprise goal sincerely due to the fact that will be your ultimate aim of the analysis. After desirable perception only we can set the precise aim of evaluation that is in sync with the enterprise objective. You need to understand if the customer desires to minimize savings loss, or if they prefer to predict the rate of a commodity, etc.

3. Data Understanding: After enterprise understanding, the subsequent step is data understanding. This includes a series of all the reachable data. Here you need to intently work with the commercial enterprise group as they are certainly conscious of what information is present, what facts should be used for this commercial enterprise problem, and different information. This step includes describing the data, their structure, their relevance, their records type. Explore the information using graphical plots. Basically, extracting any data that you can get about the information through simply exploring the data.

4. Preparation of Data: This consists of steps like choosing the applicable data, integrating the data by means of merging the data sets, cleaning it, treating the lacking values through either eliminating them or imputing them, treating inaccurate data through eliminating them, additionally test for outliers the use of box plots and cope with them. Constructing new data, derive new elements from present ones. Format the data into the preferred structure, eliminate undesirable columns and features. Data preparation is the most time-consuming but arguably the most essential step in the complete existence cycle. Your model will be as accurate as your data.

5. Exploratory Data Analysis: This step includes getting some concept about the answer and elements affecting it, earlier than constructing the real model. Distribution of data inside distinctive variables of a character is explored graphically the usage of bar-graphs, Relations between distinct aspects are captured via graphical representations like scatter plots and warmth maps. Many data visualization strategies are considerably used to discover each and every characteristic individually and by means of combining them with different features.
6. Data Modeling: Data modeling is the coronary heart of data analysis. A model takes the organized data as input and gives the preferred output. This step consists of selecting the suitable kind of model, whether the problem is a classification problem, or a regression problem or a clustering problem. After deciding on the model family, amongst the number of algorithms amongst that family, we need to cautiously pick out the algorithms to put into effect and enforce them. We need to tune the hyperparameters of every model to obtain the preferred performance. We additionally need to make positive there is the right stability between overall performance and generalizability. We do no longer desire the model to study the data and operate poorly on new data.

7. Model Evaluation: Here the model is evaluated for checking if it is geared up to be deployed. The model is examined on an unseen data, evaluated on a cautiously thought out set of assessment metrics. We additionally need to make positive that the model conforms to reality. If we do not acquire a quality end result in the evaluation, we have to re-iterate the complete modelling procedure until the preferred stage of metrics is achieved. Any data science solution, a machine learning model, simply like a human, must evolve, must be capable to enhance itself with new data, adapt to a new evaluation metric. We can construct more than one model for a certain phenomenon, however, a lot of them may additionally be imperfect. The model assessment helps us select and construct an ideal model.

8. Model Deployment: The model after a rigorous assessment is at the end deployed in the preferred structure and channel. This is the last step in the data science life cycle. Each step in the data science life cycle defined above must be laboured upon carefully. If any step is performed improperly, and hence, have an effect on the subsequent step and the complete effort goes to waste. For example, if data is no longer accumulated properly, you'll lose records and you will no longer be constructing an ideal model. If information is not cleaned properly, the model will no longer work. If the model is not evaluated properly, it will fail in the actual world. Right from Business perception to model deployment, every step has to be given appropriate attention, time, and effort.

❖ OSEMN FRAMEWORK

OSEMN is a smart acronym in which each letter represents a stage of a data science project.

- ❖ O**BTAIN:** Collect data from credible sources. — A successful project often needs more than an ad hoc manual data collection process. Depending on your goals and the sort of business or marketing queries you want to handle, your data source may differ. It might come via internal channels, such as advertising performance, allowing you to improve your preparations for the next campaign, or it could come from somewhere else.

- ❖ S**CRUB:** Clean the data such that it is consistent and usable. — Data seldom arrives in the proper format for review. Rather, it frequently comprises a large number of conflicting, erroneous, or superfluous data items. In essence, much of what is gathered is merely "noise." This is where you will move your data from its present dirty form to a clean one where it can be used.

- ❖ E**XPLORE:** Look throughout the data for noteworthy patterns in statistics that stick out. — Rather, you are investigating the data using multiple methodologies in order to gain a deeper understanding of the data and its story. You're like a data detective, looking for unusual patterns or anomalies under the surface.

- **MODEL:** Make forecasts and discoveries. — You're looking for the method that best defines how known input data may be used to predict unknown output values. Modeling is the process of using a statistical technique to better understand the patterns in your data.

- **NiNTERPRET:** Improve comprehension through data. — You reflect on the questions that prompted the data processing to occur in the first place, as well as the actionable value that comes from the process of investigating and modeling the data. You will create data visualizations to help others understand the outcomes of your study, and you will assess your level of confidence in your analysis.

REFERENCES

1. Data Science for Business – Foster Provost and Tom Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
2. Python for Data Analysis – Wes McKinney, *Python for Data Analysis*, 2nd Edition, O'Reilly Media, 2017.
3. Doing Data Science – Cathy O'Neil and Rachel Schutt, *Doing Data Science*, O'Reilly Media, 2013.
4. Data Science from Scratch – Joel Grus, *Data Science from Scratch*, 2nd Edition, O'Reilly Media, 2019.
5. Big Data Fundamentals – Thomas Erl, Wajid Khattak, and Paul Buhler, *Big Data Fundamentals*, Prentice Hall, 2015.
6. MEDIUM:"osemn framework", https://medium.com

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